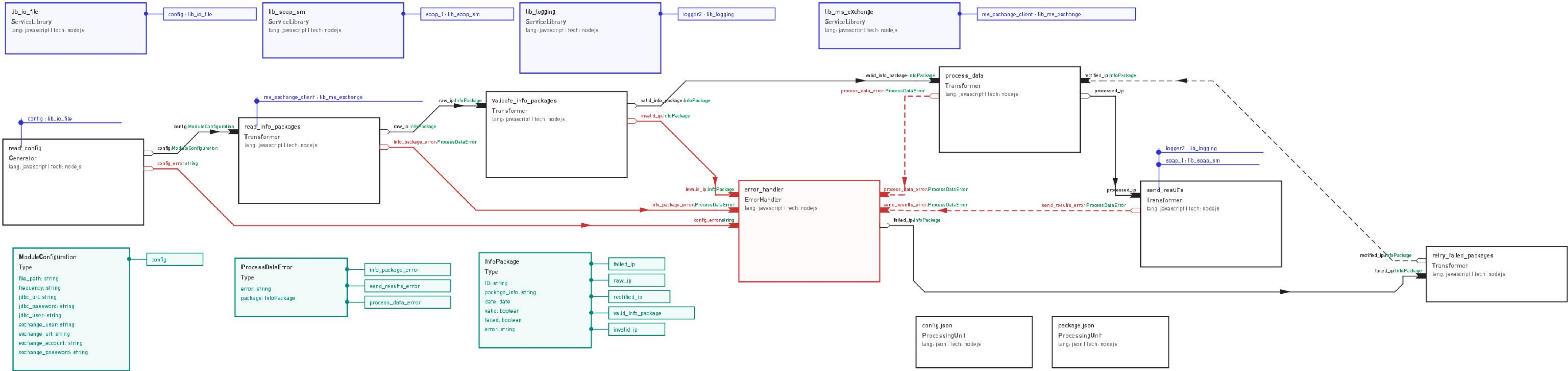


Parts List						
No.	Part	Kind	Modified	Rev.	Responsible	Signature
1	config.json	ProcessingUnit			none	
2	error_handler	ErrorHandler	2026-08-27T01:44:05Z		none	
3	InfoPackage	Type	2026-08-26T00:29:46Z		none	
4	lib_io_file	ServiceLibrary			none	
5	lib_logging	ServiceLibrary	2026-08-27T00:55:23Z		none	
6	lib_ms_exchange	ServiceLibrary	2026-08-27T01:32:55Z		none	
7	lib_soap_sm	ServiceLibrary	2026-08-27T01:29:48Z		none	
8	ModuleConfiguration	Type	2026-08-26T09:04:07Z		none	
9	package.json	ProcessingUnit			none	
10	process_data	Transformer	2026-08-27T02:01:22Z		none	
11	ProcessDataError	Type	2026-08-26T00:53:02Z		none	
12	read_config	Generator	2026-08-27T00:55:23Z		none	
13	read_info_packages	Transformer	2026-08-27T00:59:31Z		none	
14	retry_failed_packages	Transformer	2026-08-26T00:36:34Z		none	
15	send_results	Transformer	2026-08-26T00:59:12Z		none	
16	validate_info_packages	Transformer	2026-08-27T00:55:59Z		none	



Project	TestProject	Format: A1-landscape
Revision		Scale 1:1
version 2026-08-29 16:01:57		

TestProject Functional and Technical Specification

Generated from Threadwork functional and technical specifications for /TestProject.

Contents

- TestProject
- read_config
- read_info_packages
- send_results
- lib_io_file
- lib_soap_sm
- lib_logging
- error_handler
- retry_failed_packages
- config.json
- package.json
- process_data
- validate_info_packages
- lib_ms_exchange
- Shared Types
- Service Libraries

Processing Nodes

TestProject

- **Path:** /TestProject
- **Role:** CompositeWorker
- **Direct language:** javascript
- **Direct technology:** nodejs

Direct Children

Name	Stereotype
config.json	ProcessingUnit
error_handler	ErrorHandler
InfoPackage	Type
lib_io_file	ServiceLibrary
lib_logging	ServiceLibrary
lib_ms_exchange	ServiceLibrary
lib_soap_sm	ServiceLibrary
ModuleConfiguration	Type
package.json	ProcessingUnit
process_data	Transformer
ProcessDataError	Type
read_config	Generator
read_info_packages	Transformer
retry_failed_packages	Transformer
send_results	Transformer
validate_info_packages	Transformer

Flow Diagram (Mermaid)

```
flowchart LR
    node_node_0d0a832e_4daa_4f24_b58e_9010b0de3e0a["config.json (ProcessingUnit)"]
    node_node_17["error_handler (ErrorHandler)"]
    node_node_47df1883_4e55_4c08_9957_118e1e13ddde["InfoPackage (Type)"]
    node_node_5["lib_io_file (ServiceLibrary)"]
    node_node_7["lib_logging (ServiceLibrary)"]
    node_node_2bf442bf_31f6_4378_86dc_b25323b10bba["lib_ms_exchange (ServiceLibrary)"]
    node_node_6["lib_soap_sm (ServiceLibrary)"]
    node_node_464b7184_8fd2_4e9c_a5a8_2bc1e2030f45["ModuleConfiguration (Type)"]
    node_node_eceb56e4_3081_4bb3_bc18_b50a8d4e4bf0["package.json (ProcessingUnit)"]
    node_node_28["process_data (Transformer)"]
    node_node_9fc8c2eb_538e_4697_9103_91985df0ba1f["ProcessDataError (Type)"]
    node_node_1["read_config (Generator)"]
    node_node_3["read_info_packages (Transformer)"]
    node_node_22["retry_failed_packages (Transformer)"]
    node_node_4["send_results (Transformer)"]
    node_node_27["validate_info_packages (Transformer)"]
    node_node_1 -->|config| node_node_3
    node_node_5 -->|config| node_node_1
    node_node_1 -->|config_error| node_node_17
    node_node_17 -->|failed_ip| node_node_22
    node_node_3 -->|info_package_error| node_node_17
    node_node_27 -->|invalid_ip| node_node_17
    node_node_7 -->|logger2| node_node_4
    node_node_2bf442bf_31f6_4378_86dc_b25323b10bba -->|ms_exchange_client| node_node_3
    node_node_28 -->|process_data_error| node_node_17
    node_node_28 -->|processed_ip| node_node_4
    node_node_3 -->|raw_ip| node_node_27
    node_node_22 -->|rectified_ip| node_node_28
    node_node_4 -->|send_results_error| node_node_17
    node_node_6 -->|soap_1| node_node_4
    node_node_27 -->|valid_info_package| node_node_28
```

Specification

No functional specification provided.

Inputs

None.

Outputs

None.

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

read_config

- Path: /TestProject/read_config
- Role: Generator
- Direct language: Not specified directly.
- Direct technology: Not specified directly.

Specification

reads the configuration form the file config.json that is transformed into a hash map

Inputs

None.

Outputs

Name	Type	Target	Port
config	ModuleConfiguration	read_info_packages	out
config_error	string	error_handler	out

Dependencies

Instance	Library	Classification
config	lib_io_file	UsagelImport

Usage Instructions

No usage instructions provided.

Test Data

[]

read_info_packages

- **Path:** /TestProject/read_info_packages
- **Role:** Transformer
- **Direct language:** Not specified directly.
- **Direct technology:** Not specified directly.

Specification

using the credentials provided in the configuration it connects to the api endpoint specified under exchange.* it connects to ms exchange and downloads all the csv attachments in the emails from the folder Inbox/PRC/TODO puts them in the local folder todo then it opens each file transforms it into a list of InfoPackage where the package payload is the row and pushes each of the packages to raw_ip malformed rows will be pushed to info_package_error

Purpose

Retrieve CSV attachments from a configured Microsoft Exchange mailbox, convert each CSV row into an InfoPackage, and route successfully parsed packages and malformed rows to their respective outputs.

Behavior

1. Receive a ModuleConfiguration through the config input.
2. Use the Exchange connection settings from the configuration:
 - exchange_url
 - exchange_user
 - exchange_account
 - exchange_password
3. Connect to Microsoft Exchange using ms_exchange_client.
4. Access the mailbox folder:

Inbox/PRC/TODO
5. Retrieve the emails in that folder and download every CSV attachment.
6. Store the downloaded CSV files in the local folder:

todo
7. Process every downloaded CSV file.
8. For each CSV data row:
 - parse the row;
 - create one InfoPackage;
 - use the row contents as the package payload in InfoPackage.package_info;
 - emit the resulting package through raw_ip.

9. If a CSV row is malformed or cannot be transformed into a valid `InfoPackage`:
- do not emit it through `raw_ip`;
 - create a `ProcessDataError` describing the failure;
 - emit the error through `info_package_error`.

10. Processing of a malformed row must not prevent subsequent rows or files from being processed.

The specification does not define how the remaining `InfoPackage` fields (`ID`, `date`, `valid`, `failed`, and `error`) are populated. Their values must therefore follow the project's existing `InfoPackage` conventions and must not be inferred from this component specification.

Constraints

- Exchange credentials and endpoint information must come from the received `ModuleConfiguration`.
- Only CSV attachments are downloaded for processing.
- Attachments are retrieved from `Inbox/PRC/TODO`.
- Downloaded CSV files are stored in the local `todo` folder before processing.
- Each CSV data row is processed independently.
- One successfully processed row produces one `InfoPackage`.
- Malformed rows are routed to `info_package_error` and must not be routed to `raw_ip`.
- A malformed row must not terminate processing of the remaining rows or files.

Acceptance Criteria

- Given valid Exchange configuration, the component can connect to the configured Microsoft Exchange account.
- The component reads messages from `Inbox/PRC/TODO`.
- Every CSV attachment found in the selected messages is downloaded to the local `todo` folder.
- Every successfully parsed CSV data row results in exactly one `InfoPackage` emitted through `raw_ip`.
- The CSV row is represented in the emitted package's `package_info`.
- A malformed or unprocessable row produces a `ProcessDataError` through `info_package_error`.
- A malformed row does not produce an `InfoPackage` through `raw_ip`.
- Processing continues after an individual malformed row.
- The output types and link contracts remain compatible with `remove_duplicates` and `error_handler`.

Inputs

Name	Type	Source	Port
config	ModuleConfiguration	read_config	in

Outputs

Name	Type	Target	Port
info_package_error	ProcessDataError	error_handler	out
raw_ip	InfoPackage	validate_info_packages	out

Dependencies

Instance	Library	Classification
ms_exchange_client	lib_ms_exchange	UsagelImport

Usage Instructions

No usage instructions provided.

Test Data

```
{
  "component": "/TestProject/read_info_packages",
  "language": "json",
  "tests": [
    {
      "name": "process_single_csv_attachment",
      "input": {
        "config": {
          "file_path": "./todo",
          "frequency": "PT5M",
          "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEDB1",
          "jdbc_password": "test_password",
          "jdbc_user": "test_user",
          "exchange_user": "svc_prc",
          "exchange_url": "https://exchange.example.test/EWS/Exchange.asmx"
        },
        "exchange_account": "[prc@example.test](mailto:prc@example.test)",

```



```

    "exchange_password": "exchange_test_password"
  },
  "mocks": {
    "ms_exchange_client": {
      "expected_connection": {
        "url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.asmx)",
        "user": "svc_prc",
        "account": "[prc@example.test](mailto:prc@example.test)",
        "password": "exchange_test_password"
      },
      "expected_folder": "Inbox/PRC/TOD0",
      "emails": [
        {
          "subject": "PRC import",
          "attachments": [
            {
              "name": "packages.csv",
              "content_type": "text/csv",
              "content": "ID,NAME,VALUE\n1001,Alpha,42\n1002,Beta,84\n"
            }
          ]
        }
      ]
    }
  },
  "expected_files": [
    {
      "path": "todo/packages.csv",
      "exists": true
    }
  ],
  "expected_outputs": {
    "raw_ip": [
      {
        "package_info": "1001,Alpha,42"
      },
      {
        "package_info": "1002,Beta,84"
      }
    ],
    "info_package_error": []
  },
  "assertions": {
    "raw_ip_count": 2,
    "info_package_error_count": 0,
    "ignore_unspecified_InfoPackage_fields": [
      "ID",
      "date",
      "valid",
      "failed",
      "error"
    ]
  }
},
{
  "name": "process_multiple_csv_attachments_and_ignore_non_csv",
  "input": {
    "config": {
      "file_path": "./todo",
      "frequency": "PT5M",
      "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEPDB1",
      "jdbc_password": "test_password",
      "jdbc_user": "test_user",
      "exchange_user": "svc_prc",
      "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.asmx)",
      "exchange_account": "[prc@example.test](mailto:prc@example.test)",
      "exchange_password": "exchange_test_password"
    }
  },
  "mocks": {
    "ms_exchange_client": {
      "expected_folder": "Inbox/PRC/TOD0",
      "emails": [
        {
          "subject": "Batch A",
          "attachments": [
            {
              "name": "batch_a.csv",
              "content_type": "text/csv",
              "content": "ID,NAME,VALUE\n2001,Gamma,10\n2002,Delta,20\n"
            },
            {
              "name": "instructions.txt",
              "content_type": "text/plain",
              "content": "This attachment must not be processed."
            }
          ]
        }
      ],
      {
        "subject": "Batch B",
        "attachments": [
          {
            "name": "batch_b.csv",
            "content_type": "text/csv",
            "content": "ID,NAME,VALUE\n2003,Epsilon,30\n"
          }
        ]
      }
    ]
  }
}

```

```

    }
  ]
},
"expected_files": [
  {
    "path": "todo/batch_a.csv",
    "exists": true
  },
  {
    "path": "todo/batch_b.csv",
    "exists": true
  },
  {
    "path": "todo/instructions.txt",
    "exists": false
  }
],
"expected_outputs": {
  "raw_ip": [
    {
      "package_info": "2001, Gamma, 10"
    },
    {
      "package_info": "2002, Delta, 20"
    },
    {
      "package_info": "2003, Epsilon, 30"
    }
  ],
  "info_package_error": []
},
"assertions": {
  "raw_ip_count": 3,
  "info_package_error_count": 0,
  "csv_attachments_downloaded": 2,
  "non_csv_attachments_downloaded": 0,
  "ignore_unspecified_InfoPackage_fields": [
    "ID",
    "date",
    "valid",
    "failed",
    "error"
  ]
}
},
{
  "name": "route_malformed_row_to_error_and_continue",
  "input": {
    "config": {
      "file_path": "./todo",
      "frequency": "PT5M",
      "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEDB1",
      "jdbc_password": "test_password",
      "jdbc_user": "test_user",
      "exchange_user": "svc_prc",
      "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.asmx)",
      "exchange_account": "[prc@example.test](mailto:prc@example.test)",
      "exchange_password": "exchange_test_password"
    }
  },
  "mocks": {
    "ms_exchange_client": {
      "expected_folder": "Inbox/PRC/TOD0",
      "emails": [
        {
          "subject": "Mixed batch",
          "attachments": [
            {
              "name": "mixed.csv",
              "content_type": "text/csv",
              "content": "ID, NAME, VALUE\n3001, Zeta, 11\n"3002, Eta, broken\n3003, Theta, 33\n"
            }
          ]
        }
      ]
    }
  },
  "expected_outputs": {
    "raw_ip": [
      {
        "package_info": "3001, Zeta, 11"
      },
      {
        "package_info": "3003, Theta, 33"
      }
    ],
    "info_package_error": [
      {
        "error": {
          "matches": "non-empty"
        },
        "package": {
          "package_info": {
            "matches": "contains:3002"
          }
        }
      }
    ]
  }
}
}

```

```

    ]
  },
  "assertions": {
    "raw_ip_count": 2,
    "info_package_error_count": 1,
    "malformed_row_not_emitted_to_raw_ip": true,
    "processing_continues_after_malformed_row": true,
    "ignore_unspecified_InfoPackage_fields": [
      "ID",
      "date",
      "valid",
      "failed",
      "error"
    ]
  }
},
{
  "name": "empty_csv_produces_no_packages",
  "input": {
    "config": {
      "file_path": "./todo",
      "frequency": "PT5M",
      "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEPDB1",
      "jdbc_password": "test_password",
      "jdbc_user": "test_user",
      "exchange_user": "svc_prc",
      "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.
asmx)",
      "exchange_account": "[prc@example.test](mailto:prc@example.test)",
      "exchange_password": "exchange_test_password"
    }
  },
  "mocks": {
    "ms_exchange_client": {
      "expected_folder": "Inbox/PRC/TODO",
      "emails": [
        {
          "subject": "Empty batch",
          "attachments": [
            {
              "name": "empty.csv",
              "content_type": "text/csv",
              "content": "ID,NAME,VALUE\n"
            }
          ]
        }
      ]
    }
  },
  "expected_files": [
    {
      "path": "todo/empty.csv",
      "exists": true
    }
  ],
  "expected_outputs": {
    "raw_ip": [],
    "info_package_error": []
  },
  "assertions": {
    "raw_ip_count": 0,
    "info_package_error_count": 0
  }
},
{
  "name": "no_matching_email_attachments",
  "input": {
    "config": {
      "file_path": "./todo",
      "frequency": "PT5M",
      "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEPDB1",
      "jdbc_password": "test_password",
      "jdbc_user": "test_user",
      "exchange_user": "svc_prc",
      "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.
asmx)",
      "exchange_account": "[prc@example.test](mailto:prc@example.test)",
      "exchange_password": "exchange_test_password"
    }
  },
  "mocks": {
    "ms_exchange_client": {
      "expected_folder": "Inbox/PRC/TODO",
      "emails": []
    }
  },
  "expected_outputs": {
    "raw_ip": [],
    "info_package_error": []
  },
  "assertions": {
    "raw_ip_count": 0,
    "info_package_error_count": 0,
    "downloaded_csv_count": 0
  }
}
]
}

```


send_results

- **Path:** /TestProject/send_results
- **Role:** Transformer
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

Name	Type	Source	Port
processed_ip	Unspecified	process_data	in

Outputs

Name	Type	Target	Port
send_results_error	ProcessDataError	error_handler	out

Dependencies

Instance	Library	Classification
logger2	lib_logging	UsagelImport
soap_1	lib_soap_sm	UsagelImport

Usage Instructions

No usage instructions provided.

Test Data

```
[
  {
    "case": "valid",
    "input.processing_result": "{\\"id\\":\\"123\\"}",
    "expected.send_results_error": "{}"
  },
  {
    "case": "invalid",
    "input.processing_result": "{}",
    "expected.send_results_error": "{\\"error\\":\\"invalid id\\"}"
  }
]
```

lib_io_file

- **Path:** /TestProject/lib_io_file
- **Role:** ServiceLibrary
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

None.

Outputs

None.

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

lib_soap_sm

- Path: /TestProject/lib_soap_sm
- Role: ServiceLibrary
- Direct language: Not specified directly.
- Direct technology: Not specified directly.

Specification

No functional specification provided.

Inputs

None.

Outputs

None.

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

lib_logging

- Path: /TestProject/lib_logging
- Role: ServiceLibrary
- Direct language: Not specified directly.
- Direct technology: Not specified directly.

Specification

No functional specification provided.

Inputs

None.

Outputs

None.

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

error_handler

- Path: /TestProject/error_handler
- Role: ErrorHandler
- Direct language: javascript
- Direct technology: Not specified directly.

Specification

No functional specification provided.

Inputs

Name	Type	Source	Port
config_error	string	read_config	in
info_package_error	ProcessDataError	read_info_packages	in
invalid_ip	InfoPackage	validate_info_packages	in
process_data_error	ProcessDataError	process_data	in
send_results_error	ProcessDataError	send_results	in

Outputs

Name	Type	Target	Port
failed_ip	InfoPackage	retry_failed_packages	out

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

retry_failed_packages

- Path: /TestProject/retry_failed_packages
- Role: Transformer
- Direct language: Not specified directly.
- Direct technology: Not specified directly.

Specification

No functional specification provided.

Inputs

Name	Type	Source	Port
failed_ip	InfoPackage	error_handler	in

Outputs

Name	Type	Target	Port
rectified_ip	InfoPackage	process_data	out

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

```
[
  {
    "case": "valid",
    "input.failed_info_package": "{\\"id\\":\\"123\\"}",
    "expected.fixed_info_package": "{\\"id\\":\\"123\\",\\"processed\\":true}"
  }
]
```

config.json

- Path: /TestProject/config.json
- Role: ProcessingUnit
- Direct language: json
- Direct technology: Not specified directly.

Specification

No functional specification provided.

Inputs

None.

Outputs

None.

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

package.json

- Path: /TestProject/package.json
- Role: ProcessingUnit
- Direct language: json
- Direct technology: nodejs

Specification

No functional specification provided.

Inputs

None.

Outputs

None.

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

process_data

- Path: /TestProject/process_data
- Role: Transformer
- Direct language: Not specified directly.
- Direct technology: Not specified directly.

Specification

processes validInfoPackage

Inputs

Name	Type	Source	Port
rectified_ip	InfoPackage	retry_failed_packages	in
valid_info_package	InfoPackage	validate_info_packages	in

Outputs

Name	Type	Target	Port
process_data_error	ProcessDataError	error_handler	out
processed_ip	Unspecified	send_results	out

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

validate_info_packages

- Path: /TestProject/validate_info_packages
- Role: Transformer
- Direct language: Not specified directly.
- Direct technology: Not specified directly.

Specification

No functional specification provided.

Inputs

Name	Type	Source	Port
raw_ip	InfoPackage	read_info_packages	in

Outputs

Name	Type	Target	Port
invalid_ip	InfoPackage	error_handler	out
valid_info_package	InfoPackage	process_data	out

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

lib_ms_exchange

- Path: /TestProject/lib_ms_exchange
- Role: ServiceLibrary
- Direct language: Not specified directly.
- Direct technology: Not specified directly.

Specification

connects to a microsoft exchange endpoint account email using credentials and reads emails using filters like

folder:\${folderName}

Inputs

None.

Outputs

None.

Dependencies

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

Annexes

Shared Types

- InfoPackage
- ModuleConfiguration
- ProcessDataError

Service Libraries

- lib_io_file at /TestProject/lib_io_file
- lib_logging at /TestProject/lib_logging
- lib_ms_exchange at /TestProject/lib_ms_exchange
- lib_soap_sm at /TestProject/lib_soap_sm

TestProject Component Dossier

Generated from Threadwork component specifications for /TestProject.

Contents

- TestProject
- read_config
- read_info_packages
- send_results
- lib_io_file
- lib_soap_sm
- lib_logging
- error_handler
- retry_failed_packages
- config.json
- package.json
- process_data
- validate_info_packages
- lib_ms_exchange

TestProject

- Path: /TestProject
- Role: CompositeWorker
- Direct language: javascript
- Direct technology: nodejs

Direct Children

Name	Stereotype
config.json	ProcessingUnit
error_handler	ErrorHandler
InfoPackage	Type
lib_io_file	ServiceLibrary
lib_logging	ServiceLibrary
lib_ms_exchange	ServiceLibrary
lib_soap_sm	ServiceLibrary
ModuleConfiguration	Type
package.json	ProcessingUnit
process_data	Transformer
ProcessDataError	Type
read_config	Generator
read_info_packages	Transformer
retry_failed_packages	Transformer
send_results	Transformer
validate_info_packages	Transformer

Flow Diagram (Mermaid)

```
flowchart LR
    node_node_0d0a832e_4daa_4f24_b58e_9010b0de3e0a["config.json (ProcessingUnit)"]
    node_node_17["error_handler (ErrorHandler)"]
    node_node_47df1883_4e55_4c08_9957_118e1e13ddde["InfoPackage (Type)"]
    node_node_5["lib_io_file (ServiceLibrary)"]
    node_node_7["lib_logging (ServiceLibrary)"]
    node_node_2bf442bf_31f6_4378_86dc_b25323b10bba["lib_ms_exchange (ServiceLibrary)"]
    node_node_6["lib_soap_sm (ServiceLibrary)"]
    node_node_464b7184_8fd2_4e9c_a5a8_2bc1e2030f45["ModuleConfiguration (Type)"]
    node_node_eceb56e4_3081_4bb3_bc18_b50a8d4e4bf0["package.json (ProcessingUnit)"]
    node_node_28["process_data (Transformer)"]
    node_node_9fc8c2eb_538e_4697_9103_91985df0ba1f["ProcessDataError (Type)"]
    node_node_1["read_config (Generator)"]
    node_node_3["read_info_packages (Transformer)"]
    node_node_22["retry_failed_packages (Transformer)"]
    node_node_4["send_results (Transformer)"]
    node_node_27["validate_info_packages (Transformer)"]
    node_node_1 -->|config| node_node_3
    node_node_5 -->|config| node_node_1
    node_node_1 -->|config_error| node_node_17
    node_node_17 -->|failed_ip| node_node_22
    node_node_3 -->|info_package_error| node_node_17
    node_node_27 -->|invalid_ip| node_node_17
    node_node_7 -->|logger2| node_node_4
    node_node_2bf442bf_31f6_4378_86dc_b25323b10bba -->|ms_exchange_client| node_node_3
    node_node_28 -->|process_data_error| node_node_17
    node_node_28 -->|processed_ip| node_node_4
    node_node_3 -->|raw_ip| node_node_27
    node_node_22 -->|rectified_ip| node_node_28
    node_node_4 -->|send_results_error| node_node_17
    node_node_6 -->|soap_1| node_node_4
    node_node_27 -->|valid_info_package| node_node_28
```

Specification

No functional specification provided.

Inputs

None.

Incoming Link Contracts

None.

Outputs

None.

Outgoing Link Contracts

None.

Used Types

None.

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

read_config

- **Path:** /TestProject/read_config
- **Role:** Generator
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

reads the configuration form the file config.json that is transformed into a hash map

Inputs

None.

Incoming Link Contracts

None.

Outputs

Name	Type	Target	Port
config	ModuleConfiguration	read_info_packages	out
config_error	string	error_handler	out

Outgoing Link Contracts

config:ModuleConfiguration

- **Link ID:** link_14
- **From:** read_config.out
- **To:** read_info_packages.in
- **Transport:** packet
- **Classification:** Transport

Shared type: ModuleConfiguration

Field	Type	Reference
file_path	string	No
frequency	string	No
jdbc_url	string	No
jdbc_password	string	No
jdbc_user	string	No
exchange_user	string	No
exchange_url	string	No
exchange_account	string	No
exchange_password	string	No

config_error:string

- **Link ID:** link_18
- **From:** read_config.out
- **To:** error_handler.in
- **Transport:** error
- **Classification:** ErrorPipe

No payload type definition provided.

Used Types

ModuleConfiguration

Field	Type	Reference
file_path	string	No
frequency	string	No
jdbc_url	string	No
jdbc_password	string	No
jdbc_user	string	No
exchange_user	string	No
exchange_url	string	No
exchange_account	string	No
exchange_password	string	No

Used Libraries

config

- **Library:** lib_io_file
- **Library path:** /TestProject/lib_io_file
- **Classification:** UsageImport
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Usage Instructions

No usage instructions provided.

Test Data

[]

read_info_packages

- **Path:** `/TestProject/read_info_packages`
- **Role:** Transformer
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

using the credentials provided in the configuration it connects to the api endpoint specified under exchange.* it connects to ms exchange and downloads all the csv attachments in the emails from the folder Inbox/PRC/TODO puts them in the local folder todo then it opens each file transforms it into a list of `InfoPackage` where the package payload is the row and pushes each of the packages to `raw_ip` malformed rows will be pushed to `info_package_error`

Purpose

Retrieve CSV attachments from a configured Microsoft Exchange mailbox, convert each CSV row into an `InfoPackage`, and route successfully parsed packages and malformed rows to their respective outputs.

Behavior

1. Receive a `ModuleConfiguration` through the `config` input.
2. Use the Exchange connection settings from the configuration:
 - `exchange_url`
 - `exchange_user`
 - `exchange_account`
 - `exchange_password`
3. Connect to Microsoft Exchange using `ms_exchange_client`.
4. Access the mailbox folder:
`Inbox/PRC/TODO`
5. Retrieve the emails in that folder and download every CSV attachment.
6. Store the downloaded CSV files in the local folder:
`todo`
7. Process every downloaded CSV file.
8. For each CSV data row:
 - parse the row;
 - create one `InfoPackage`;
 - use the row contents as the package payload in `InfoPackage.package_info`;
 - emit the resulting package through `raw_ip`.
9. If a CSV row is malformed or cannot be transformed into a valid `InfoPackage`:
 - do not emit it through `raw_ip`;
 - create a `ProcessDataError` describing the failure;
 - emit the error through `info_package_error`.
10. Processing of a malformed row must not prevent subsequent rows or files from being processed.

The specification does not define how the remaining `InfoPackage` fields (`ID`, `date`, `valid`, `failed`, and `error`) are populated. Their values must therefore follow the project's existing `InfoPackage` conventions and must not be inferred from this component specification.

Constraints

- Exchange credentials and endpoint information must come from the received `ModuleConfiguration`.
- Only CSV attachments are downloaded for processing.
- Attachments are retrieved from `Inbox/PRC/TODO`.
- Downloaded CSV files are stored in the local `todo` folder before processing.
- Each CSV data row is processed independently.
- One successfully processed row produces one `InfoPackage`.
- Malformed rows are routed to `info_package_error` and must not be routed to `raw_ip`.
- A malformed row must not terminate processing of the remaining rows or files.

Acceptance Criteria

- Given valid Exchange configuration, the component can connect to the configured Microsoft Exchange account.
- The component reads messages from `Inbox/PRC/TODO`.
- Every CSV attachment found in the selected messages is downloaded to the local `todo` folder.
- Every successfully parsed CSV data row results in exactly one `InfoPackage` emitted through `raw_ip`.
- The CSV row is represented in the emitted package's `package_info`.
- A malformed or unprocessable row produces a `ProcessDataError` through `info_package_error`.
- A malformed row does not produce an `InfoPackage` through `raw_ip`.
- Processing continues after an individual malformed row.
- The output types and link contracts remain compatible with `remove_duplicates` and `error_handler`.

Inputs

Name	Type	Source	Port
config	ModuleConfiguration	read_config	in

Incoming Link Contracts

config:ModuleConfiguration

- **Link ID:** `link_14`
- **From:** `read_config.out`
- **To:** `read_info_packages.in`
- **Transport:** `packet`
- **Classification:** `Transport`

Shared type: `ModuleConfiguration`

Field	Type	Reference
file_path	string	No
frequency	string	No
jdbc_url	string	No
jdbc_password	string	No
jdbc_user	string	No
exchange_user	string	No
exchange_url	string	No
exchange_account	string	No
exchange_password	string	No

Outputs

Name	Type	Target	Port
info_package_error	ProcessDataError	error_handler	out
raw_ip	InfoPackage	validate_info_packages	out

Outgoing Link Contracts

info_package_error:ProcessDataError

- **Link ID:** `link_19`
- **From:** `read_info_packages.out`
- **To:** `error_handler.in`
- **Transport:** `error`
- **Classification:** `ErrorPipe`

Shared type: `ProcessDataError`

Field	Type	Reference
error	string	No
package	InfoPackage	No

raw_ip:InfoPackage

- **Link ID:** `link_89bc0a9e-04c3-4100-8267-0be2f650db9f`
- **From:** `read_info_packages.out`
- **To:** `validate_info_packages.in`
- **Transport:** `in-process`
- **Classification:** `Transport`

Shared type: `InfoPackage`

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Used Types

ModuleConfiguration

Field	Type	Reference
file_path	string	No
frequency	string	No
jdbc_url	string	No
jdbc_password	string	No
jdbc_user	string	No
exchange_user	string	No
exchange_url	string	No
exchange_account	string	No
exchange_password	string	No

ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Used Libraries

ms_exchange_client

- **Library:** `lib_ms_exchange`

- **Library path:** /TestProject/lib_ms_exchange
- **Classification:** UsageImport
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Library specification

connects to a microsoft exchange endpoint account email using credentials and reads emails using filters like

folder:\${folderName}

Usage Instructions

No usage instructions provided.

Test Data

```
{
  "component": "/TestProject/read_info_packages",
  "language": "json",
  "tests": [
    {
      "name": "process_single_csv_attachment",
      "input": {
        "config": {
          "file_path": "./todo",
          "frequency": "PT5M",
          "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEDB1",
          "jdbc_password": "test_password",
          "jdbc_user": "test_user",
          "exchange_user": "svc_prc",
          "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.asmx)",
          "exchange_account": "[prc@example.test](mailto:prc@example.test)",
          "exchange_password": "exchange_test_password"
        }
      },
      "mocks": {
        "ms_exchange_client": {
          "expected_connection": {
            "url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.asmx)",
            "user": "svc_prc",
            "account": "[prc@example.test](mailto:prc@example.test)",
            "password": "exchange_test_password"
          },
          "expected_folder": "Inbox/PRC/TODO",
          "emails": [
            {
              "subject": "PRC import",
              "attachments": [
                {
                  "name": "packages.csv",
                  "content_type": "text/csv",
                  "content": "ID,NAME,VALUE\n1001,Alpha,42\n1002,Beta,84\n"
                }
              ]
            }
          ]
        }
      },
      "expected_files": [
        {
          "path": "todo/packages.csv",
          "exists": true
        }
      ],
      "expected_outputs": {
        "raw_ip": [
          {
            "package_info": "1001,Alpha,42"
          },
          {
            "package_info": "1002,Beta,84"
          }
        ],
        "info_package_error": []
      },
      "assertions": {
        "raw_ip_count": 2,
        "info_package_error_count": 0,
        "ignore_unspecified_InfoPackage_fields": [
          "ID",
          "date",
          "valid",
          "failed",
          "error"
        ]
      }
    },
    {
      "name": "process_multiple_csv_attachments_and_ignore_non_csv",
      "input": {
        "config": {

```



```

        "file_path": "./todo",
        "frequency": "PT5M",
        "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEPDB1",
        "jdbc_password": "test_password",
        "jdbc_user": "test_user",
        "exchange_user": "svc_prc",
        "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.
asmx)",
        "exchange_account": "[prc@example.test](mailto:prc@example.test)",
        "exchange_password": "exchange_test_password"
    },
    "mocks": {
        "ms_exchange_client": {
            "expected_folder": "Inbox/PRC/TOD0",
            "emails": [
                {
                    "subject": "Batch A",
                    "attachments": [
                        {
                            "name": "batch_a.csv",
                            "content_type": "text/csv",
                            "content": "ID,NAME,VALUE\n2001, Gamma, 10\n2002, Delta, 20\n"
                        },
                        {
                            "name": "instructions.txt",
                            "content_type": "text/plain",
                            "content": "This attachment must not be processed."
                        }
                    ]
                },
                {
                    "subject": "Batch B",
                    "attachments": [
                        {
                            "name": "batch_b.csv",
                            "content_type": "text/csv",
                            "content": "ID,NAME,VALUE\n2003, Epsilon, 30\n"
                        }
                    ]
                }
            ]
        }
    },
    "expected_files": [
        {
            "path": "todo/batch_a.csv",
            "exists": true
        },
        {
            "path": "todo/batch_b.csv",
            "exists": true
        },
        {
            "path": "todo/instructions.txt",
            "exists": false
        }
    ],
    "expected_outputs": {
        "raw_ip": [
            {
                "package_info": "2001, Gamma, 10"
            },
            {
                "package_info": "2002, Delta, 20"
            },
            {
                "package_info": "2003, Epsilon, 30"
            }
        ],
        "info_package_error": []
    },
    "assertions": {
        "raw_ip_count": 3,
        "info_package_error_count": 0,
        "csv_attachments_downloaded": 2,
        "non_csv_attachments_downloaded": 0,
        "ignore_unspecified_InfoPackage_fields": [
            "ID",
            "date",
            "valid",
            "failed",
            "error"
        ]
    },
    {
        "name": "route_malformed_row_to_error_and_continue",
        "input": {
            "config": {
                "file_path": "./todo",
                "frequency": "PT5M",
                "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEPDB1",
                "jdbc_password": "test_password",
                "jdbc_user": "test_user",
                "exchange_user": "svc_prc",
                "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.
asmx)",
                "exchange_account": "[prc@example.test](mailto:prc@example.test)",

```

```

        "exchange_password": "exchange_test_password"
    },
    "mocks": {
        "ms_exchange_client": {
            "expected_folder": "Inbox/PRC/TOD0",
            "emails": [
                {
                    "subject": "Mixed batch",
                    "attachments": [
                        {
                            "name": "mixed.csv",
                            "content_type": "text/csv",
                            "content": "ID,NAME,VALUE\n3001,Zeta,11\n"3002,Eta,broken\n3003,Theta,33\n"
                        }
                    ]
                }
            ]
        }
    },
    "expected_outputs": {
        "raw_ip": [
            {
                "package_info": "3001,Zeta,11"
            },
            {
                "package_info": "3003,Theta,33"
            }
        ],
        "info_package_error": [
            {
                "error": {
                    "matches": "non-empty"
                },
                "package": {
                    "package_info": {
                        "matches": "contains:3002"
                    }
                }
            }
        ]
    },
    "assertions": {
        "raw_ip_count": 2,
        "info_package_error_count": 1,
        "malformed_row_not_emitted_to_raw_ip": true,
        "processing_continues_after_malformed_row": true,
        "ignore_unspecified_InfoPackage_fields": [
            "ID",
            "date",
            "valid",
            "failed",
            "error"
        ]
    },
    {
        "name": "empty_csv_produces_no_packages",
        "input": {
            "config": {
                "file_path": "./todo",
                "frequency": "PT5M",
                "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEDB1",
                "jdbc_password": "test_password",
                "jdbc_user": "test_user",
                "exchange_user": "svc_prc",
                "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.
asmx)",
                "exchange_account": "[prc@example.test](mailto:prc@example.test)",
                "exchange_password": "exchange_test_password"
            }
        },
        "mocks": {
            "ms_exchange_client": {
                "expected_folder": "Inbox/PRC/TOD0",
                "emails": [
                    {
                        "subject": "Empty batch",
                        "attachments": [
                            {
                                "name": "empty.csv",
                                "content_type": "text/csv",
                                "content": "ID,NAME,VALUE\n"
                            }
                        ]
                    }
                ]
            }
        },
        "expected_files": [
            {
                "path": "todo/empty.csv",
                "exists": true
            }
        ],
        "expected_outputs": {
            "raw_ip": [],
            "info_package_error": []
        }
    },

```

```
    "assertions": {
      "raw_ip_count": 0,
      "info_package_error_count": 0
    },
  },
  {
    "name": "no_matching_email_attachments",
    "input": {
      "config": {
        "file_path": "./todo",
        "frequency": "PT5M",
        "jdbc_url": "jdbc:oracle:thin:@localhost:1521/FREEDB1",
        "jdbc_password": "test_password",
        "jdbc_user": "test_user",
        "exchange_user": "svc_prc",
        "exchange_url": "[https://exchange.example.test/EWS/Exchange.asmx](https://exchange.example.test/EWS/Exchange.
asmx)",
        "exchange_account": "[prc@example.test](mailto:prc@example.test)",
        "exchange_password": "exchange_test_password"
      }
    },
    "mocks": {
      "ms_exchange_client": {
        "expected_folder": "Inbox/PRC/TODO",
        "emails": []
      }
    },
    "expected_outputs": {
      "raw_ip": [],
      "info_package_error": []
    },
    "assertions": {
      "raw_ip_count": 0,
      "info_package_error_count": 0,
      "downloaded_csv_count": 0
    }
  }
]
}
```

send_results

- **Path:** /TestProject/send_results
- **Role:** Transformer
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

Name	Type	Source	Port
processed_ip	Unspecified	process_data	in

Incoming Link Contracts

processed_ip

- **Link ID:** link_6e7c73cc-a766-4cb6-aeb9-715a6476c0d1
- **From:** process_data.out
- **To:** send_results.in
- **Transport:** in-process
- **Classification:** Transport

No payload type definition provided.

Outputs

Name	Type	Target	Port
send_results_error	ProcessDataError	error_handler	out

Outgoing Link Contracts

send_results_error:ProcessDataError

- **Link ID:** link_21
- **From:** send_results.out
- **To:** error_handler.in
- **Transport:** error
- **Classification:** ErrorPipe

Shared type: ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

Used Types

ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

Used Libraries

logger2

- **Library:** lib_logging
- **Library path:** /TestProject/lib_logging
- **Classification:** UsageImport
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

soap_1

- **Library:** lib_soap_sm
- **Library path:** /TestProject/lib_soap_sm
- **Classification:** UsageImport
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Usage Instructions

No usage instructions provided.

Test Data

```
[
  {
    "case": "valid",
    "input.processing_result": "{\id\:\123\}",
    "expected.send_results_error": "{}"
  },
  {
    "case": "invalid",
    "input.processing_result": "{}",
    "expected.send_results_error": "{\error\:\invalid id\}"
  }
]
```

lib_io_file

- **Path:** /TestProject/lib_io_file
- **Role:** ServiceLibrary
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

None.

Incoming Link Contracts

None.

Outputs

None.

Outgoing Link Contracts

None.

Used Types

None.

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

lib_soap_sm

- **Path:** /TestProject/lib_soap_sm
- **Role:** ServiceLibrary
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

None.

Incoming Link Contracts

None.

Outputs

None.

Outgoing Link Contracts

None.

Used Types

None.

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

lib_logging

- **Path:** /TestProject/lib_logging
- **Role:** ServiceLibrary
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

None.

Incoming Link Contracts

None.

Outputs

None.

Outgoing Link Contracts

None.

Used Types

None.

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

error_handler

- **Path:** /TestProject/error_handler
- **Role:** ErrorHandler
- **Direct language:** javascript
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

Name	Type	Source	Port
config_error	string	read_config	in
info_package_error	ProcessDataError	read_info_packages	in
invalid_ip	InfoPackage	validate_info_packages	in
process_data_error	ProcessDataError	process_data	in
send_results_error	ProcessDataError	send_results	in

Incoming Link Contracts

config_error:string

- **Link ID:** link_18
- **From:** read_config.out
- **To:** error_handler.in
- **Transport:** error
- **Classification:** ErrorPipe

No payload type definition provided.

info_package_error:ProcessDataError

- **Link ID:** link_19
- **From:** read_info_packages.out
- **To:** error_handler.in
- **Transport:** error
- **Classification:** ErrorPipe

Shared type: ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

invalid_ip:InfoPackage

- **Link ID:** link_738b0fd6-d53d-4c2d-a8b3-ba8730b2725b
- **From:** validate_info_packages.out
- **To:** error_handler.in
- **Transport:** in-process
- **Classification:** ErrorPipe

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

process_data_error:ProcessDataError

- **Link ID:** link_336c7a11-7dd5-46fe-a134-cbcd18dde831
- **From:** process_data.out
- **To:** error_handler.in
- **Transport:** in-process
- **Classification:** ErrorPipe

Shared type: ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

send_results_error:ProcessDataError

- **Link ID:** link_21
- **From:** send_results.out
- **To:** error_handler.in
- **Transport:** error
- **Classification:** ErrorPipe

Shared type: ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

Outputs

Name	Type	Target	Port
failed_ip	InfoPackage	retry_failed_packages	out

Outgoing Link Contracts

failed_ip:InfoPackage

- **Link ID:** link_23
- **From:** error_handler.out
- **To:** retry_failed_packages.in
- **Transport:** in-process
- **Classification:** Transport

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Used Types

ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

retry_failed_packages

- **Path:** /TestProject/retry_failed_packages
- **Role:** Transformer
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

Name	Type	Source	Port
failed_ip	InfoPackage	error_handler	in

Incoming Link Contracts

failed_ip:InfoPackage

- **Link ID:** link_23
- **From:** error_handler.out
- **To:** retry_failed_packages.in
- **Transport:** in-process
- **Classification:** Transport

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Outputs

Name	Type	Target	Port
rectified_ip	InfoPackage	process_data	out

Outgoing Link Contracts

rectified_ip:InfoPackage

- **Link ID:** link_066ebc49-c6b5-44fa-ba93-e2b7029c9aab
- **From:** retry_failed_packages.out
- **To:** process_data.in
- **Transport:** in-process
- **Classification:** Transport

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Used Types

InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

```
[
  {
    "case": "valid",
    "input.failed_info_package": "{\"id\":\"123\"}",
    "expected.fixed_info_package": "{\"id\":\"123\", \"processed\":true}"
  }
]
```

config.json

- **Path:** /TestProject/config.json
- **Role:** ProcessingUnit
- **Direct language:** json
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

None.

Incoming Link Contracts

None.

Outputs

None.

Outgoing Link Contracts

None.

Used Types

None.

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

package.json

- Path: /TestProject/package.json
- Role: ProcessingUnit
- Direct language: json
- Direct technology: nodejs

Specification

No functional specification provided.

Inputs

None.

Incoming Link Contracts

None.

Outputs

None.

Outgoing Link Contracts

None.

Used Types

None.

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

process_data

- **Path:** /TestProject/process_data
- **Role:** Transformer
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

processes validInfoPackage

Inputs

Name	Type	Source	Port
rectified_ip	InfoPackage	retry_failed_packages	in
valid_info_package	InfoPackage	validate_info_packages	in

Incoming Link Contracts

rectified_ip:InfoPackage

- **Link ID:** link_066ebc49-c6b5-44fa-ba93-e2b7029c9aab
- **From:** retry_failed_packages.out
- **To:** process_data.in
- **Transport:** in-process
- **Classification:** Transport

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

valid_info_package:InfoPackage

- **Link ID:** link_30
- **From:** validate_info_packages.out
- **To:** process_data.in
- **Transport:** in-process
- **Classification:** Transport

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Outputs

Name	Type	Target	Port
process_data_error	ProcessDataError	error_handler	out
processed_ip	Unspecified	send_results	out

Outgoing Link Contracts

process_data_error:ProcessDataError

- **Link ID:** link_336c7a11-7dd5-46fe-a134-cbcd18dde831
- **From:** process_data.out
- **To:** error_handler.in
- **Transport:** in-process
- **Classification:** ErrorPipe

Shared type: ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

processed_ip

- **Link ID:** link_6e7c73cc-a766-4cb6-aeb9-715a6476c0d1
- **From:** process_data.out
- **To:** send_results.in
- **Transport:** in-process
- **Classification:** Transport

No payload type definition provided.

Used Types

InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

ProcessDataError

Field	Type	Reference
error	string	No
package	InfoPackage	No

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

validate_info_packages

- **Path:** /TestProject/validate_info_packages
- **Role:** Transformer
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

No functional specification provided.

Inputs

Name	Type	Source	Port
raw_ip	InfoPackage	read_info_packages	in

Incoming Link Contracts

raw_ip:InfoPackage

- **Link ID:** link_89bc0a9e-04c3-4100-8267-0be2f650db9f
- **From:** read_info_packages.out
- **To:** validate_info_packages.in
- **Transport:** in-process
- **Classification:** Transport

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Outputs

Name	Type	Target	Port
invalid_ip	InfoPackage	error_handler	out
valid_info_package	InfoPackage	process_data	out

Outgoing Link Contracts

invalid_ip:InfoPackage

- **Link ID:** link_738b0fd6-d53d-4c2d-a8b3-ba8730b2725b
- **From:** validate_info_packages.out
- **To:** error_handler.in
- **Transport:** in-process
- **Classification:** ErrorPipe

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

valid_info_package:InfoPackage

- **Link ID:** link_30
- **From:** validate_info_packages.out
- **To:** process_data.in
- **Transport:** in-process
- **Classification:** Transport

Shared type: InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Used Types

InfoPackage

Field	Type	Reference
ID	string	No
package_info	string	No
date	date	No
valid	boolean	No
failed	boolean	No
error	string	No

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.

lib_ms_exchange

- **Path:** /TestProject/lib_ms_exchange
- **Role:** ServiceLibrary
- **Direct language:** *Not specified directly.*
- **Direct technology:** *Not specified directly.*

Specification

connects to a microsoft exchange endpoint account email using credentials and reads emails using filters like

folder:\${folderName}

Inputs

None.

Incoming Link Contracts

None.

Outputs

None.

Outgoing Link Contracts

None.

Used Types

None.

Used Libraries

None.

Usage Instructions

No usage instructions provided.

Test Data

No test data provided.